

A common zero-free strip for Hecke L -functions

AI-assisted research draft*

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Abstract

We prove a zero-free strip of width $25/258$ for all finite-order Hecke L -functions, conditional on the cited theta and reflection inputs and the fixed-order large sieve. The proof first treats Dedekind zeta functions. Each reflected row then pairs a character with its conjugate, and one symmetric zero-free rectangle controls both L -factors. The required density estimate retains the number of selected rows in a fourth L -moment and a second mollifier moment. Its exponent is determined by a quadratic equation, and a quartic inequality gives the stated rational width. The same argument reaches an exactly defined endpoint $0.096906264531643757\dots$. The width is common to all fixed fields; uniform implied constants in a varying field are not asserted.

1 The theorem and the reflection input

We take as analytic inputs the theta realization, reflected expansion, and their estimates in [1, Sections 3–8 and Appendix A]. The construction there is presented over cyclotomic fields. We first verify the fixed-field generality needed here and record the precise consequences used in the proof. The large-sieve input is stated separately in Section 2. These cited analytic results are retained as inputs; their full proofs are not reproduced or independently certified in this article.

Theorem 1.1. *Under the theta and reflection inputs in Proposition 1.2 and the fixed-order large sieve (2.1), for every number field K and every finite-order Hecke character η of K ,*

$$L_K(s, \eta) \neq 0 \quad \left(\Re s > \frac{233}{258} \right),$$

with the usual pole at $s = 1$ permitted for the principal character. More precisely, the conclusion holds in $\Re s > 1 - \Delta_\#$, where $\Delta_\# > 25/258$ is defined in (4.2); its numerical value is $0.096906264531643757\dots$

All implied constants may depend on the fixed field, the local data, and fixed auxiliary parameters. No zero-free hypothesis is part of the inputs. The new estimates below are deduced from the displayed identities and the large sieve.

Proposition 1.2 (Fixed-field theta reflection). *Under the cited theta and reflection inputs, the construction of [1] applies to every fixed number field $F \supset \mu_{12}$ after fixing a finite set S and its local data. It supplies sixth-power-free rows u , characters χ_u of order dividing six, and a probe $I(X, Y, Z)$ with the following properties.*

The row norm is $q_u = Nu$, the actual primitive conductor satisfies $Q_u \ll_F q_u$, and the principal inducing row is $u = 1$. After a fixed finite ray-class partition, the characters form the

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residue-character family in [2]. For $X = Y = Z^{1-h}$, $1/2 < h < 3/5$, the reflected row square and the additive square mass satisfy

$$\sum_m |B_m(Z)|^2 \ll Z^\varepsilon (R_0 + R_0^2/Z), \quad R_0 = XY, \quad \sum_m |A_m|^2 \ll X^{-1} + Y^{-1}, \quad (1.1)$$

and $|I| \leq (\sum |A_m|^2 \sum |B_m|^2)^{1/2}$.

For the principal target, the Poisson expansion is a sum of triple Mellin integrals with scalar factor

$$X^{1/2-z} Z^{\xi+z-1} Y^{w-1} q_u^{-z} \quad (1.2)$$

and row Dirichlet series

$$\zeta_F^S(6z) \frac{L_F^S(w, \chi_u)}{L_F^S(\xi, \bar{\chi}_u)} H_u(\xi, w, z). \quad (1.3)$$

The local factors are those in [1, (7.16)–(7.20)]. The fixed Mellin weights in w, z are rapidly decreasing on vertical lines, with the Gaussian factor $e^{(\xi+z-1)^2}$. The double principal residue at $w = 1, z = 1/6$ is $c_0 f_h(Z)$, where $c_0 \neq 0$ and

$$f_h(Z) = \frac{1}{2\pi i} \int_{(2)} Z^{s-1+(1-h)/2+h/6} e^{(s-5/6)^2} \frac{H_1(s)}{\zeta_F^S(s)} ds, \quad H_1(s) = H_1(s, 1, 1/6). \quad (1.4)$$

The expansions and contour truncations have the convergence and vertical decay asserted in [1, Sections 7–8].

Proof. Enlarge S to contain the places above six, the trace-character ramification, and primes generating the ideal class group. The ring of S -integers is then principal. The Poisson lattice in [1, Lemma 3.1] uses adelic duality, the product formula, and strong approximation over F . Balanced generators follow from compactness of the norm-one idele class group. Write $[F : \mathbb{Q}] = 2r$; all its archimedean places are complex. For $\ell = 2, 3$, the local and global S -unit power-class spaces in [1, Lemma 3.2] have dimensions

$$\dim_{\mathbb{F}_\ell}(F_S^\times / F_S^{\times \ell}) = 2|S_f| + 2r, \quad \dim_{\mathbb{F}_\ell}(U_S / U_S^\ell) = |S_f| + r.$$

The trivial S -class group gives injectivity of the global image; Hilbert reciprocity makes it isotropic. Since -1 is a sixth power in F , the pairing is alternating. Thus the image is Lagrangian and has the isotropic complement required by the construction. Its Kummer characters and conductor bounds use only $\mu_6 \subset F$ and class field theory.

For the cubic Gauss correction, put $E_0 = \mathbb{Q}(\sqrt{-3})$ and $\Lambda = \Lambda_{E_0} \circ N_{F/E_0}$. If $\mathfrak{p} \mid \mathfrak{q}$ has residue degree f , Jacobi-sum lifting gives $J_{\kappa(\mathfrak{p})} = (-1)^{f-1} J_{\kappa(\mathfrak{q})}^f = -\pi^+(\mathfrak{q})^f$. This proves the identity $\gamma_2(\mathfrak{p})^3 = -\Lambda(\mathfrak{p})$ used in [1, Lemma 3.4]. At every complex place the relative norm contributes the coordinate or its conjugate, so the angular exponent is $+1$ or -1 , with no radial part. This supplies the constant-term cancellation even when F is not CM. The quadratic correction uses the Weil-index product formula and the same isotropic complement.

The metaplectic framework of [4, Section 0.2 and Chapter II] is over any global field containing the required roots of unity. For the specialization (rank, cover degree, c) = (2, 3, 2) in [1, Proposition 5.1 and Appendix A], the scalar center, exceptional quotient, spherical vector, and local Whittaker conditions depend on the local fields and these parameters. The adelic splitting uses Hilbert reciprocity. Compatibility of the global datum with rational lifts uses that an everywhere-local cube is a global cube, which follows from Kummer theory and Chebotarev. The cited nonzero theta residue and Whittaker functional have the same general-field scope.

All archimedean profiles are therefore the complex profiles used in the source. The ensuing reflection uses those profiles and finite power-class partitions; its ideal counts and quadratic large sieve hold over each fixed number field. The separated additive fractions use balanced generators

and the product formula, and the exterior local Gauss identities use $N\mathfrak{p} \equiv 1 \pmod{12}$. No step here requires $F/\mathbb{Q}$ to be Galois or abelian.

To justify the changed probe scales, retain the dyads in the proof of [1, Section 6.4], before its final use of a fixed row-box exponent. Let $P = N\mathfrak{h}$ denote the powerful part and $D = N\mathfrak{d}$ the squarefree part. The compact row box gives $PD \ll R_0 = XY$. The local reflection and quadratic large sieve give, with the source's row restrictions retained,

$$\sum_{\mathfrak{h} \sim P} \sum_{\mathfrak{d} \sim D} |B_{hd}(Z)|^2 \ll Z^\varepsilon P^{1/2} \left(D + \frac{D^2 P}{Z} \right) \ll Z^\varepsilon \left(\frac{R_0}{\sqrt{P}} + \frac{R_0^2}{Z\sqrt{P}} \right).$$

Summing dyads proves the first estimate in (1.1). The retained column cutoff is $Z^\nu(8D^2P/Z)$; the decay order is chosen after ν in [1, (6.40)–(6.45)]. If this cutoff is below one, the full partial-sum estimate (6.40) is already a rapidly decaying tail, so no constant term is added. The additive square norm in [1, Section 7.1] is $(XY + Y^2)/(XY^2) = X^{-1} + Y^{-1}$. The Poisson identities and the two principal residues then give (1.2)–(1.4) with these scales. All constants depend only on the fixed field and auxiliary data. $\square$

1.1 The local Euler domain

The exact local sums give a domain larger than the sufficient domain recorded in [1, (7.21)]. This enlargement is needed for the lines used in the proof.

Lemma 1.3. *The correction H_u is holomorphic and $\ll q_u^\varepsilon$, uniformly in the imaginary parts, on every closed subregion with positive margins of*

$$\begin{aligned} \Re \xi > \tfrac{1}{2}, \quad \Re w > 0, \quad \Re z > 0, \quad \Re(\xi + w) > 1, \\ \Re(w + 6z) > 1, \quad \Re(\xi + 6z) > 1, \quad \Re(\xi + w + 6z) > 2, \quad \Re(\xi + z) > \tfrac{5}{6}. \end{aligned} \tag{1.5}$$

After a fixed enlargement of S , $H_1(s)$ is nonzero for $\Re s \geq 9/10$.

Proof. Write $p = N\mathfrak{p}$, and use real parts for the exponents in the following bounds. At a prime not dividing u , the exact factor in [1, (7.19)] differs from one by at most a constant times

$$p^{-w-6z} + p^{-\xi-6z} + p^{-\xi-w} + p^{1-\xi-w-6z} + p^{4-6\xi-6z}.$$

Every term is $O(p^{-1-\delta})$ on a fixed inner region of (1.5). At a prime with $j = v_{\mathfrak{p}}(u)$, $1 \leq j \leq 5$, the exact boundary terms instead give

$$\begin{aligned} |H_{u,\mathfrak{p}} - 1| &\ll p^{4-6\xi-6z} + p^{1-\xi-w} + \mathbf{1}_{j=5} p^{3-6\xi} + \mathbf{1}_{j=2} p^{3/2-3\xi} \\ &\quad + \mathbf{1}_{j=3} (p^{2-3\xi-w} + p^{2-4\xi}) + \mathbf{1}_{j=4} p^{5/2-4\xi-w}. \end{aligned}$$

To obtain this bound directly from [1, (7.17)], write the theta exponent as $t = 6a, 6a + 1, 6a + 3, 6a + 4$. For the other modulus exponents 0, 1, the boundary terms occur at $j + 6m = t - 1, t$, respectively. Away from these boundaries the character sums vanish unless $t \equiv 0, 1 \pmod{6}$, respectively. The remaining terms sum geometrically with ratio of absolute value $p^{4-6\xi-6z}$; the displayed bound retains each boundary. All its exponents are negative on (1.5). Thus the good-prime product converges normally and the bad-prime product is bounded by $\prod_{\mathfrak{p}|u} (1 + Cp^{-\delta}) \ll q_u^\varepsilon$. For $u = 1$ every exterior prime is good. Enlarge S so that each remaining correction has distance less than $1/2$ from one on $\Re s \geq 9/10$, $w = 1$, $z = 1/6$; the convergent logarithmic product proves nonvanishing. $\square$

2 Moments on selected rows

Fix F and its local data. We suppress the superscript S in moment estimates and omit the principal inducing row. The input from [2, Theorem 1.1], with character order six, is

$$\sum_{qu \leq U} \left| \sum_{\mathbf{Nn} \leq V} a_{\mathbf{n}} \chi_u(\mathbf{n}) \right|^2 \ll (UV)^\varepsilon \mathcal{K}(U, V) \sum |a_{\mathbf{n}}|^2, \quad \mathcal{K}(U, V) = U + V + U^{5/6} V^{1/3} + U^{1/3} V^{5/6}, \quad (2.1)$$

initially for sixth-power-free coefficient ideals. Fixed twists, ray-class partitions and row subsets are permitted.

We will also use (2.1) for divisor-bounded coefficients on arbitrary ideals. First restrict the complete ideal to $\mathbf{Nn} \asymp V$ and write $\mathbf{n} = \mathbf{b}\mathbf{a}^6$. On a real line s , its squared coefficient mass for fixed $\mathbf{a}$ is $O(V^{1-2s+\varepsilon}/(\mathbf{N}\mathbf{a})^6)$. Minkowski's inequality and the convergent sum in $(\mathbf{N}\mathbf{a})^{-3+\varepsilon}$ give

$$\sum_{qu \leq U} |D_u(s + it; V)|^2 \ll (UV)^\varepsilon \mathcal{K}(U, V) V^{1-2s}. \quad (2.2)$$

Here $\chi_u(\mathbf{a})^6$ merely restricts to rows coprime to $\mathbf{a}$. Smooth weights can be separated on conductor and column dyads by Mellin inversion.

The approximate functional equation and (2.2) give the three full-family bounds needed later:

$$\begin{aligned} \sum |L(1/2 + it, \chi_u)|^2 &\ll U^{1+\varepsilon} T_0^C, & \sum |L(1/2 + it, \chi_u)|^4 &\ll U^{7/6+\varepsilon} T_0^C, \\ \sum |L(1/4 + it, \chi_u)|^8 &\ll U^{3+\varepsilon} T_0^C, \end{aligned} \quad (2.3)$$

where $T_0 = 2 + |t|$ and C is fixed. Indeed, apply the approximate functional equation to L^m , with $m = 1, 2, 4$, on actual conductor dyads $Q \ll U$; the polynomial lengths are $Q^{m/2} T_0^C$. Maximizing the four terms of (2.2) at their two column endpoints gives the displayed powers. On the dual side retain the squared multiplier $Q^{m(1-2s)}$ until the endpoints have been evaluated. Its lower endpoint is at most the displayed bound; its upper endpoint has the same conductor exponent as the original polynomial. Conjugation inside an absolute square restores the residue-character family, and the root number is a row phase of modulus one.

For $A \geq 1$, put

$$M_A(s, \chi_u) = \sum_{\mathbf{Nn} \leq A} \mu_F(\mathbf{n}) \chi_u(\mathbf{n}) (\mathbf{Nn})^{-s}.$$

The next lemma preserves the size of an arbitrary selected set in both factors of the zero detector.

Lemma 2.1. *Let $\mathcal{R}$ be a set of $R \geq 1$ rows of norm at most U , and set $c = 2 - 2\tau$, where $7/12 \leq \tau \leq 2/3$. At a common height t ,*

$$\sum_{u \in \mathcal{R}} |L(\tau + it, \chi_u)|^4 \ll U^\varepsilon T_0^C \left(U^c R^{1-c} + U^{c+1/6} \right), \quad (2.4)$$

$$\sum_{u \in \mathcal{R}} |M_A(\tau + it, \chi_u)|^2 \ll (UA)^\varepsilon \left(U^c R^{1-c} + A^c + U^{5/6} A^{c-2/3} + U^{1/3} A^{c-1/6} \right), \quad (2.5)$$

where (2.5) requires $H = U/R \leq A$. Fixed multiplicative constants or subpower factors in H are harmless.

Proof. The part of a divisor-bounded polynomial of norm at most H is $O(H^{1-\tau+\varepsilon})$ in every row. Its squared sum on $\mathcal{R}$ is $RH^c = U^c R^{1-c}$, up to allowed losses. For a column dyad $V > H$, (2.2) is $\mathcal{K}(U, V) V^{c-1}$. Its row term decreases with V , and its other slopes $c, c - 2/3, c - 1/6$ are nonnegative. Evaluating at H and A proves (2.5).

For (2.4), apply the approximate functional equation to L^2 and first partition the actual conductor into dyads Q . Ignore polynomial height factors temporarily; both polynomials have

length Q . For the main polynomial, $Q \leq H$ gives the rowwise estimate $RQ^c \leq U^c R^{1-c}$. If $Q > H$, split at H as above. Its other endpoint terms are at most U^c , $U^{5/6}U^{c-2/3}$ and $U^{1/3}U^{c-1/6}$, all bounded by $U^{c+1/6}$.

The dual multiplier has absolute size Q^{c-1} before squaring, and the dual coefficient line is $1 - \tau = c/2$. All column slopes on that line are positive. At its upper endpoint, after the squared multiplier is inserted, the four large-sieve terms are

$$UQ^{c-1}, \quad Q^c, \quad U^{5/6}Q^{c-2/3}, \quad U^{1/3}Q^{c-1/6}. \quad (2.6)$$

If $Q \geq H$, the first is at most $UH^{c-1} = U^c R^{1-c}$ because its conductor exponent is negative; the others are at most $U^{c+1/6}$. If $Q < H$, the entire dual polynomial, including its multiplier, has squared rowwise bound RQ^c . This is again at most $U^c R^{1-c}$. Thus no negative conductor power has been replaced prematurely by a power of U . The smooth weights, sixth-power removal and dual conjugation are handled as in (2.2); they cost only the displayed polynomial height and subpower factors. $\square$

3 The density estimate

For $T \geq 1$, let $R(\sigma, T; U)$ count rows whose inducing nonprincipal L -function has a zero in $\Re s \geq \sigma$, $|\Im s| \leq T$. We first state the detector in a form that retains the selected set.

Lemma 3.1. *Choose $1 \leq A < B$, $1/2 \leq \tau < \sigma < 1$, with $\sigma - \tau$ bounded below. Apart from factors $(UABT)^\varepsilon T^C$, the polynomial-tail alternative for the selected rows is bounded by*

$$B_\sigma = B^{2-2\sigma} + UA^{1-2\sigma} + \sup_{A \leq V \leq B} V^{1-2\sigma} (U^{1/3}V^{5/6} + U^{5/6}V^{1/3}). \quad (3.1)$$

If R selected rows satisfy the integral alternative, then

$$R^3 \ll (UABT)^\varepsilon T^C B^{4(\tau-\sigma)} \sup_{|v| \leq T+C \log(UABT+3)} \left(\sum_{\mathcal{R}} |L(\tau + iv)|^4 \right) \left(\sum_{\mathcal{R}} |M_A(\tau + iv)|^2 \right)^2. \quad (3.2)$$

Proof. Select one zero $\varrho_u = \beta_u + i\gamma_u$ in each row. Mellin inversion and a shift to $\Re z = \tau - \beta_u$ give

$$e^{-1/B} + \sum_{N\mathbf{n} > A} a_A(\mathbf{n}) \chi_u(\mathbf{n}) (N\mathbf{n})^{-\varrho_u} e^{-N\mathbf{n}/B} = \frac{1}{2\pi i} \int_{(\tau-\beta_u)} L(\varrho_u + z) M_A(\varrho_u + z) \Gamma(z) B^z dz,$$

where $a_A(\mathbf{n}) = \sum_{\mathfrak{d}|\mathbf{n}, N\mathfrak{d} \leq A} \mu_F(\mathfrak{d})$. The crossed Gamma residue vanishes at ϱ_u ; the omitted principal row is the only possible L -pole. Each row therefore has a large integral or a large tail. Gamma decay truncates the integral at logarithmic distance from γ_u . Substituting $v = \gamma_u + \Im z$ puts all rows on a common height integral, with a uniform envelope because $\beta_u - \tau$ stays away from zero and one. Summing and applying Hölder with exponents 4, 2, 4 gives

$$R \ll (UABT)^\varepsilon T^C B^{\tau-\sigma} R^{1/4} \sup_v \left(\sum_{\mathcal{R}} |L|^4 \right)^{1/4} \left(\sum_{\mathcal{R}} |M_A|^2 \right)^{1/2},$$

which proves (3.2).

For the tail, dyadic subdivision and partial summation remove the selected β_u . The one-dimensional Sobolev inequality replaces each row-dependent height supremum by an integral of $|D_u|^2 + |D'_u|^2$ on a slightly larger interval. Binary blocks handle the partial sums; derivatives and blocks cost logarithms. At a common ordinate apply (2.2). Maximizing its two pure terms at A, B gives (3.1); the mixed terms are left as a supremum. This is the detector argument of [3, Section 5], with the kernel (2.1). The exponentially decaying part beyond B changes only allowed losses. At least half the rows satisfy one of the alternatives. $\square$

Proposition 3.2. For $1/6 \leq d \leq 1/3$, let $x = x(d)$ be the unique positive root of

$$\Phi_d(x) = 6d(6d^2 - 15d + 7)x^2 + (54d^2 - 39d + 4)x + 2d - 2 = 0. \quad (3.3)$$

Then, for some fixed C ,

$$R(2/3 + d/2, T; U) \ll U^{k(d)+\varepsilon} T^C, \quad k(d) = \frac{5}{6} - dx(d). \quad (3.4)$$

Proof. The quadratic coefficient is positive and the constant is negative. The evaluations

$$\Phi_d(5/7) = \frac{(6d-1)(150d^2 - 35d - 42)}{49} \leq 0, \quad \Phi_d(1) = 1 + (1-3d)(1+8d-12d^2) \geq 1$$

give $5/7 \leq x < 1$. Put

$$y = \frac{1-2dx}{1-2d}, \quad D_0 = x(1-d) - \frac{1}{6}, \quad c = \frac{x(2/3-d)}{D_0}, \quad \tau = 1 - \frac{c}{2}. \quad (3.5)$$

Here $D_0 \geq 13/42$, and

$$c - \frac{2}{3} = \frac{1-3dx}{9D_0} \geq 0, \quad \frac{5}{6} - c = \frac{6x(1+d) - 5}{36D_0} \geq 0.$$

Also $1 \leq y \leq 11/7$, $x \leq y$, and $\sigma - \tau \geq 1/12$ for $\sigma = 2/3 + d/2$.

Take $A = U^x$, $B = U^y$ and write $q = 2\sigma - 1 = d + 1/3$. The two mixed terms in (3.1) have exponents $1/3 + (5/6 - q)y$ and $5/6 + (1/3 - q)x$, both equal to k . The differences from the other two exponents $1 - qx$ and $(1 - q)y$ are $(2x - 1)/6$ and $(2 - y)/6$, both nonnegative.

At $R = U^k$, the condition $H \leq A$ follows from $D_0 > 0$. The row term in (2.5) balances its first mixed term because

$$c + (1 - c)k = \frac{5}{6} + (c - 2/3)x. \quad (3.6)$$

Its gaps from the other two terms are $5/6 - 2x/3$ and $(1 - x)/2$. The row term in (2.4) is no larger than $U^{c+1/6}$: indeed

$$D_0 - (2x - 1)k = 2(1 - x)(1/3 - dx) \geq 0 \implies (1 - c)k \leq 1/6.$$

The exponent on the right of (3.2) is therefore $4(\tau - \sigma)y + (c + 1/6) + 2\{c + (1 - c)k\}$. It equals $3k$, since

$$(1 + 2c)k = 3c + \frac{1}{6} + (4/3 - 2c - 2d)y \quad (3.7)$$

reduces exactly to (3.3) after (3.5) is substituted.

This verification bounds the unknown R self-consistently. For $R \geq U^k$, the condition $H \leq A$ remains true. Relative to their values at U^k , each sparse moment grows by at most $(R/U^k)^{1-c}$. Thus (3.2) implies

$$(R/U^k)^3 \ll U^\varepsilon T^C (R/U^k)^{3(1-c)}.$$

Since $c \geq 2/3$, this gives (3.4). For $R < U^k$ the claim is immediate. The tail alternative was already bounded. $\square$

3.1 The exterior density ranges

For $0 \leq d \leq 1/2$ set

$$D = 2 + 9d - 6d^2, \quad \hat{x} = \frac{8d+1}{D}, \quad \hat{y} = \frac{11d+2}{D}, \quad \hat{k} = \frac{5}{6} - d\hat{x}. \quad (3.8)$$

Then $R(2/3 + d/2, T; U) \ll U^{\widehat{k} + \varepsilon} T^C$. To see this, use (3.2) on $\tau = 1/2$ with $A = U^{\widehat{x}}$, $B = U^{\widehat{y}}$. Here $1/2 \leq \widehat{x} \leq 1$, $1 \leq \widehat{y} \leq 3/2$. The full fourth moment is $U^{7/6}$, and the second mollifier moment is at most $U^{5/6 + \widehat{x}/3}$. After taking the cube root, the integral exponent is

$$\frac{17}{18} + \frac{2\widehat{x}}{9} - \left(\frac{2}{9} + \frac{2d}{3}\right)\widehat{y}.$$

Substitution makes it and the two mixed tail exponents equal to $\widehat{k}$; the other gaps are again $(2\widehat{x} - 1)/6$ and $(2 - \widehat{y})/6$.

For $1/2 \leq \sigma \leq 2/3$, the row count at the endpoint and [2, Corollary 1.5] give, with $q = 2\sigma - 1$,

$$R(\sigma, T; U) \ll U^{1-q/(4-6q)+\varepsilon} T^C. \quad (3.9)$$

For $11/12 \leq \sigma < 1$, we use

$$R(\sigma, T; U) \ll U^{(1-\sigma)(84\sigma-53)/6+\varepsilon} T^C. \quad (3.10)$$

For completeness, apply Hölder with exponents 4, 4, 2 in the integral alternative at $\tau = 1/2$. The fourth mollifier moment is bounded by $\mathcal{K}(U, A^2)$, so the resulting detector is

$$R \ll (UABT)^\varepsilon T^C \left(B^{1-2\sigma} U^{7/12} \mathcal{K}(U, A^2)^{1/2} + \mathcal{B}_\sigma \right).$$

Set $q = 2\sigma - 1$, $x = (7q - 3)/2$, $y = x + 7/12$, $A = U^x$, $B = U^y$. Since $q \geq 5/6$, the kernel exponent is $2x$. The integral term, the B term and the larger mixed tail term have exponent $(1 - q)(42q - 11)/12$; the other terms are smaller. This proves (3.10).

4 The density envelope and the zero bins

Use $x(d)$ from (3.3) on $[1/6, 1/3]$ and $\widehat{x}(d)$ from (3.8) on its complement in $[0, 1/2]$. Denote this piecewise choice by $x_*(d)$ and put $k = 5/6 - dx_*(d)$, $r(d) = d(1 - x_*(d))$.

Lemma 4.1. *For $2/3 \leq \alpha \leq 11/12$, with $d = 2\alpha - 4/3$,*

$$k + 2\alpha - \frac{1}{2} = \frac{5}{3} + r(d) \leq \frac{43}{25} - \frac{m_0}{101250}, \quad m_0 = \frac{247341}{71872} > 0. \quad (4.1)$$

Furthermore, if

$$r_\# = \max_{1/6 \leq d \leq 1/3} d(1 - x(d)), \quad h_\# = \frac{1}{5/3 + r_\#}, \quad \Delta_\# = \frac{h_\#}{6}, \quad (4.2)$$

then $k + 2\alpha - 1/2 \leq 1/h_\#$ throughout the same interval and $25/43 < h_\# < 57/98 < 7/12$.

Proof. For the central interval test $x_0 = 1 - 4/(75d)$. Direct substitution and $t = d - 1/4$ give

$$\begin{aligned} 75^2 d \Phi_d(x_0) &= 202500d^4 - 224100d^3 + 66501d^2 - 3690d - 528 \\ &= 202500t^4 - 21600t^3 - \frac{51273}{2}t^2 + 198t - \frac{303}{64} \\ &\leq -\frac{30321}{4}t^2 + 198t - \frac{303}{64} \leq \frac{39204}{30321} - \frac{303}{64} = -m_0, \end{aligned}$$

where $|t| \leq 1/4$ suffices for the first inequality. Thus $x_0 < x(d)$. Between these two values, $\partial_x \Phi_d < 18$; the mean value theorem gives $d(x - x_0) \geq m_0/101250$ and hence (4.1).

For the exterior parameters, the exact identities

$$\begin{aligned} \frac{1}{20} - d(1 - \widehat{x}) &= \frac{(1 - 6d)(2 + d - 20d^2)}{20D} \geq 0 \quad (0 \leq d \leq 1/6), \\ \frac{2}{39} - d(1 - \widehat{x}) &= \frac{(3d - 1)(78d^2 + 9d - 4)}{39D} \geq 0 \quad (1/3 \leq d \leq 1/2) \end{aligned}$$

give stronger bounds than (4.1). Finally, $\Phi_{1/4}(15/19) = 81/5776 > 0$ implies $r_\# > 1/19$. This exceeds both exterior bounds, so the central maximum controls the entire interval. Combining $1/19 < r_\# < 4/75$ gives the asserted range of $h_\#$. $\square$

4.1 The numerator and denominator

Fix a small grid width $\lambda > 0$, a dyad $q_u \asymp U$, and a small fixed height exponent $\nu > 0$. Put a nonprincipal row in the bin $[\alpha, \alpha + \lambda]$ when its rightmost zero in $\Re s \geq 1/2$, $|\Im s| \leq 4U^\nu$ has real part in that interval. Rows with no such zero form the floor group.

For a binned row, on the fixed real-part range used below,

$$|L_F^S(s, \chi_u)| + |L_F^S(s, \chi_u)|^{-1} \ll U^\varepsilon \quad (\Re s \geq \alpha + 2\lambda, |\Im s| \leq U^\nu). \quad (4.3)$$

Indeed choose nested discs centered at $2 + it$ whose largest disc stays to the right of $\alpha + \lambda$ and within the zero-free rectangle. A branch of $\log L$ is bounded on a small disc in absolute convergence. The polynomial conductor-height bound and Borel–Carathéodory bound it by $O(\log U)$ on a larger inner disc. Three-circles interpolation on an intermediate disc reaching $\alpha + 2\lambda + it$ gives $O((\log U)^\theta)$ with fixed $\theta < 1$. Exponentiating gives both parts of (4.3). Since $L_F^S(s, \overline{\chi_u}) = \overline{L_F^S(\bar{s}, \chi_u)}$, the symmetric rectangle gives the same bounds for $\overline{\chi_u}$, in particular the reciprocal bound for the denominator in (1.3). The same argument applies to the floor with left boundary $1/2$.

For $\alpha \leq 11/12$ choose

$$\xi = \alpha + 4\lambda, \quad w = 1 - \alpha - 2\lambda, \quad z = 1/3 + \lambda. \quad (4.4)$$

The primitive functional equation sends the numerator at $w + it$ to the conjugate character at $1 - w - it = \alpha + 2\lambda - it$. The fixed local factors at S are harmless, so (4.3) gives

$$|L_F^S(w + it, \chi_u)| \ll U^{\alpha - 1/2 + 2\lambda + \varepsilon} (2 + |t|)^C.$$

Here the conductor exponent is nonnegative, so $Q_u \ll U$ can be used directly. Choose ν small enough to absorb the height factor. A bin containing $O(U^{k+\varepsilon})$ rows therefore has absolute ratio sum bounded by

$$U^{\ell + O(\lambda) + \varepsilon}, \quad \ell = k + \alpha - \frac{1}{2}. \quad (4.5)$$

The numerator and denominator heights can vary independently in their central ranges. Both are controlled by the same zero-free rectangle; no independence of a selected set is required.

5 Contour estimates and completion of the proof

Fix $25/43 \leq h \leq h_\#$ and write $a = 1 - h$, $X = Y = Z^a$. Since $2a < 1$, (1.1) gives

$$I(Z^a, Z^a, Z) \ll Z^{a/2 + \varepsilon}. \quad (5.1)$$

For rows $U = Z^t$ whose absolute ratio sum has exponent ℓ , (1.2) gives the relative exponent

$$\mathcal{E} - a/2 = \xi - 1 + hz + a(w - 1) + t(\ell - z). \quad (5.2)$$

Take a grid containing $1/2, 2/3, 3/4, 5/6, 11/12$ and sufficiently small λ , and split at $U = Z^{1/6}$ and $U = Z^{h+\lambda}$. All intermediate rows use $z = 1/3 + \lambda$.

The following table gives the complete bounds. Lines and exponents in the table omit $O(\lambda)$ offsets; the actual bin lines are (4.4). Every bound is relative to $Z^{a/2}$.

Range	(ξ, w, z)	Ratio exponent ℓ	Relative exponent
Principal w remainder	$(1, 2/3, 1/6)$	0	$\leq -1/24$
Principal z remainder	$(1, 1, 0)$	0	0
$U \leq Z^{1/6}$	$(1, 1/100, 1/3)$	299/200	$\leq -11/450$
Floor	$(1/2, 1/2, 1/3)$	1	$\leq -1/8$
$1/2 \leq \alpha \leq 2/3$	$(\alpha, 1 - \alpha, 1/3)$	$k + \alpha - 1/2$	$\leq -1/36$
$2/3 \leq \alpha \leq 11/12$	$(\alpha, 1 - \alpha, 1/3)$	$k + \alpha - 1/2$	≤ 0
$11/12 \leq \alpha < 1$	$(\alpha, 1/4, 1/3)$	$7k/8 + 3/8$	$\leq -1/144$
$U > Z^{h+\lambda}$	$(4, 4, 6/\lambda + 1)$	1	$\leq -2h$

Here are the contour choices and verifications. For the principal row first take $\xi = 1 + \lambda$, move w to $2/3$ with $z = 1/6 + \lambda$, and take the pole at $w = 1$. Inside its residue move z to λ and take the pole at $1/6$. The first remainder has relative exponent $h/2 - 1/3 \leq -1/24$, and the second has exponent $O(\lambda)$. These are sequential moves; the pair $w = 2/3, z = \lambda$ is never required.

For small rows take $\xi = 1 + \lambda, w = 1/100$. Pointwise convexity and the row count give $\ell = 299/200$. Substitution at $t = 1/6$ gives the table, since $h \leq 7/12$. For the floor, take $\xi = 1/2 + 2\lambda, w = 1/2$ and use (4.3) and the first moment from (2.3). Its exponent at $t = h$ is $-1 + 3h/2 \leq -1/8$.

For bins below $11/12$, (5.2) and (4.5) simplify at $t = h$ to

$$-1 + h \left(k + 2\alpha - \frac{1}{2} \right). \quad (5.3)$$

In the low range put $q = 2\alpha - 1$. The function $1 - q/(4 - 6q) + q + 1/2$ increases on $[0, 1/3]$ and has maximum $5/3$, giving $-1/36$. In the middle range Lemma 4.1 makes (5.3) nonpositive for $h \leq h_{\#}$. At $h = 25/43$ it is at most $-m_0/174150 < -1/60000$.

For the highest bins take $\xi = \alpha + 2\lambda, w = 1/4$. The reciprocal is controlled by (4.3); Hölder and the eighth moment in (2.3) give $\ell = 7k/8 + 3/8$, with k from (3.10). The relative exponent at $t = h$ is $\alpha - 7/4 + h(9/8 + 7k/8)$. Its derivative in α is negative in this interval for $h \geq 25/43$. Its maximum is $-5/6 + 17h/12 \leq -1/144$.

In every intermediate group $\ell - z > 0$. In the middle bins, $\ell = 1 + d(1/2 - x_*) \geq 3/4$; in the highest bins $\ell \geq 3/8 > 1/3$. The largest dyadic endpoint is consequently $t = h + \lambda$, changing the bounds by $O(\lambda)$. All contours satisfy (1.5). On the bin contours (4.4), the potentially tight local Euler inequalities have margins $\xi + w - 1 = 2\lambda$ and $\xi + z - 5/6 \geq 5\lambda$.

For large rows move to $\xi = w = 4$ and $z = 6/\lambda + 1$. Absolute convergence gives the row exponent one. Since $z > 1$, the worst endpoint is $t = h + \lambda$; insertion in (5.2) gives $-2h$.

Finally, for each fixed grid choose ν sufficiently small that the factors T^C in the density and moment estimates, with $T = 4U^\nu$, cost only $U^{O(\lambda)}$. Truncate the original Mellin contours before shifting their central segments. The horizontal joins lie in the zero-free rectangles used for (4.3); the remaining tails stay on lines of absolute convergence. The Gaussian and rapidly decreasing Mellin weights give any fixed power saving there. Since $U > Z^{1/6}$ for intermediate rows, these savings also control the horizontal integrals in the Z aspect. The small-row and large-row contours need only absolute convergence and convexity. All fixed decay orders and local data are chosen before Z grows.

The Poisson expansion, the residues and the table now yield

$$I(Z^a, Z^a, Z) = c_0 f_h(Z) + O(Z^{a/2+C\lambda}), \quad f_h(Z) \ll Z^{a/2+C\lambda}.$$

For $0 < Z \leq 1$, moving the line in (1.4) arbitrarily far right gives arbitrary power decay. Its Mellin transform therefore continues $e^{(s-5/6)^2} H_1(s)/\zeta_F^S(s)$ holomorphically to $\Re s > 1 - h/6 + C\lambda$. By Lemma 1.3 its numerator does not vanish in the required region. Letting $\lambda \downarrow 0$ proves that ζ_F has no zeros in $\Re s > 1 - h/6$, including $h = h_{\#}$; nonpositive limiting exponents are sufficient for this open half-plane conclusion.

To finish, let K be arbitrary. The extension $K(\mu_{12})/K$ is finite abelian and its Dedekind zeta function factors into ζ_K and nonprincipal, entire Hecke factors. The strip proved for the larger field therefore excludes zeros of ζ_K away from one. Next, any finite-order Hecke character η occurs in the zeta factorization of a finite abelian extension E/K . The strip for ζ_E excludes zeros of $L_K(s, \eta)$ away from one. Comparing the nonzero simple residues of ζ_E and ζ_K gives nonvanishing of each nonprincipal factor at one. This proves Theorem 1.1.

A The optimized endpoint

Here $x(d)$ always denotes the central root in (3.3). Set $r = d(1 - x)$ and

$$Q(d, r) = d\Phi_d(1 - r/d) = 36d^4 - (36 + 72r)d^3 + (5 + 126r + 36r^2)d^2 \\ + (2 - 45r - 90r^2)d - 4r + 42r^2.$$

Along the root, $Q_r = -\partial_x \Phi_d < 0$. On $1/6 \leq d \leq 1/3$, $0 \leq r \leq 4/75$,

$$Q_{dd} = 432d^2 - (216 + 432r)d + 10 + 252r + 72r^2 \leq -\frac{2622}{625} < 0.$$

Indeed this expression increases with r and is convex in d ; its maximum is obtained by setting $r = 4/75$ and testing the two endpoints in d . At every stationary point of $r(d)$, $r'' = Q_{dd}/\partial_x \Phi_d < 0$, so there is at most one. There is an interior maximum because

$$r(1/6) = 1/21 < 1/19, \quad \Phi_{1/3}(16/19) = -28/361 < 0, \quad \Phi_{1/4}(15/19) = 81/5776 > 0.$$

Thus the maximum defining $r_{\#}$ is attained at a unique $d_{\#}$.

Its value can be enclosed using rational arithmetic alone. For rational d , bisect the positive root $x(d)$ in $[5/7, 1]$. The sign of $r'(d)$ is the sign of

$$g(d, x) = (36d^3 - 42d)x^2 + (72d^3 - 126d^2 + 84d - 4)x + 54d^2 - 37d + 4.$$

Rational interval evaluations of g permit an outer bisection of $d_{\#}$. The bounds $\partial_x \Phi_d > 3$ and $|\partial_d \Phi_d| \leq 38$ give $|r'(d)| < 6$ on this interval. Hence a root interval $[d_-, d_+]$ and an interval evaluation of r at its midpoint enclose $r_{\#}$ after adding $3(d_+ - d_-)$ to the upper endpoint. With 180 outer and 280 inner bisections, exact rational arithmetic gives

$$0.253879034133118787 < d_{\#} < 0.253879034133118788, \\ 0.053208384468374847 < r_{\#} < 0.053208384468374848, \\ 0.096906264531643757 < \Delta_{\#} < 0.096906264531643758.$$

An exact-arithmetic implementation accompanies the article. The numerical enclosure is not needed for the rational width $25/258$ or for the exact variational definition (4.2).

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